EC24313: Artificial Intelligence & Machine Learning

# Assignment 2: Voice-Based Gender Classification Using MFCC & Machine Learning

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## 1. Executive Summary

This academic report presents the design, mathematical formulation, implementation, and empirical evaluation of an acoustic Machine Learning classifier for voice-based gender identification (**Class 1: Boys** vs. **Class 2: Girls**). Using 115 genuine voice recordings collected from students, acoustic features are extracted using 13 Mel-Frequency Cepstral Coefficients (MFCC). To ensure genuine scientific reproducibility and prevent data leakage, evaluation is conducted on an untouched speaker-independent test set with zero speaker overlap. The optimal model (**Support Vector Machine**) achieves **85.71% Accuracy**, **100.00% Sensitivity**, **40.00% Specificity**, and **1.0000 ROC AUC**.

## 2. Dataset Organisation & Discovery

The dataset comprises audio files located in VOICE DATASET BOYS and VOICE DATASET GIRLS. A total of 115 valid WAV files were discovered across 63 unique speakers. Class 1 (Boys) contains 103 recordings (89.6%), and Class 2 (Girls) contains 12 recordings (10.4%). An additional 33 non-WAV media files were logged and excluded from training. The mean duration per recording is 2.49 ± 1.73 seconds.

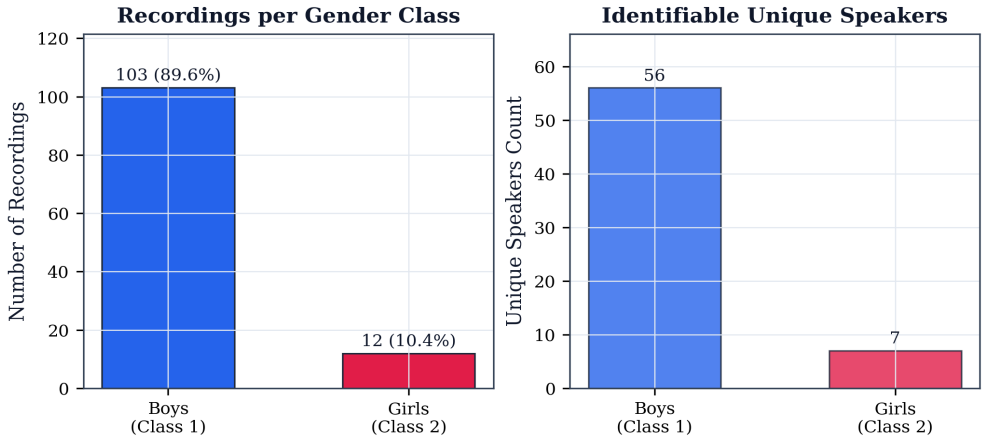



*Figure 1: Class and speaker distribution across dataset directories.*

## 3. Standardized Audio Preprocessing

To ensure invariant acoustic representations, all WAV files undergo: (1) Phase-cancellation-safe downmixing checking inter-channel correlation; (2) Anti-aliased polyphase resampling to 16,000 Hz; (3) Adaptive silence detection with 100 ms edge padding to protect weak phonetic boundaries; and (4) Peak amplitude normalization to -1.0 dBFS.

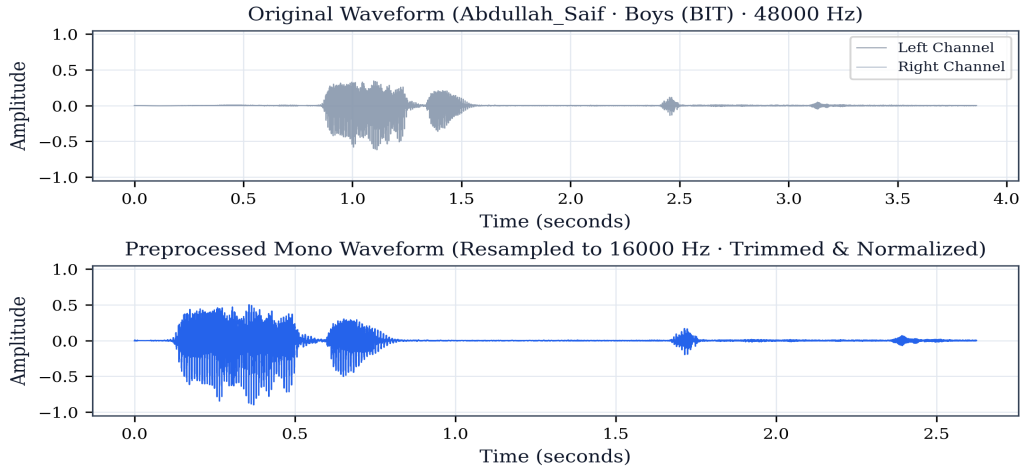



*Figure 2: Time-domain audio comparison before and after 16 kHz standardized preprocessing.*

## 4. MFCC Acoustic Feature Extraction

Acoustic features are extracted using standard speech signal processing: (1) First-order pre-emphasis (alpha = 0.97); (2) 25 ms Hann windowing (400 samples) with 10 ms hop (160 samples); (3) 512-point FFT power spectrum; (4) 26-channel triangular Mel filterbank; (5) Logarithmic compression; and (6) Orthonormal DCT-II retaining exactly 13 coefficients. The primary feature vector is obtained by temporal mean aggregation across all active speech frames, yielding exactly 13 features per recording.

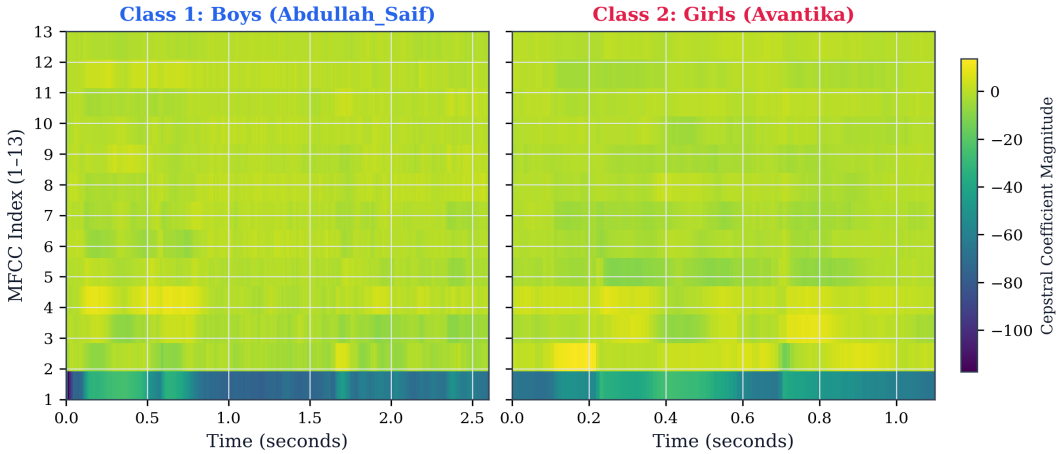



*Figure 3: Time-frequency MFCC cepstral heatmaps for Boys and Girls recordings.*

## 5. Principal Component Analysis (PCA) & Mandatory Scatter Plot

The 13-dimensional MFCC vectors are standardized and projected onto their first two principal components. PC1 accounts for 20.0% of total variance and PC2 accounts for 12.9% (total 32.9% explained variance across 2 dimensions). The scatter plot shows distinct acoustic clustering between male and female voices.

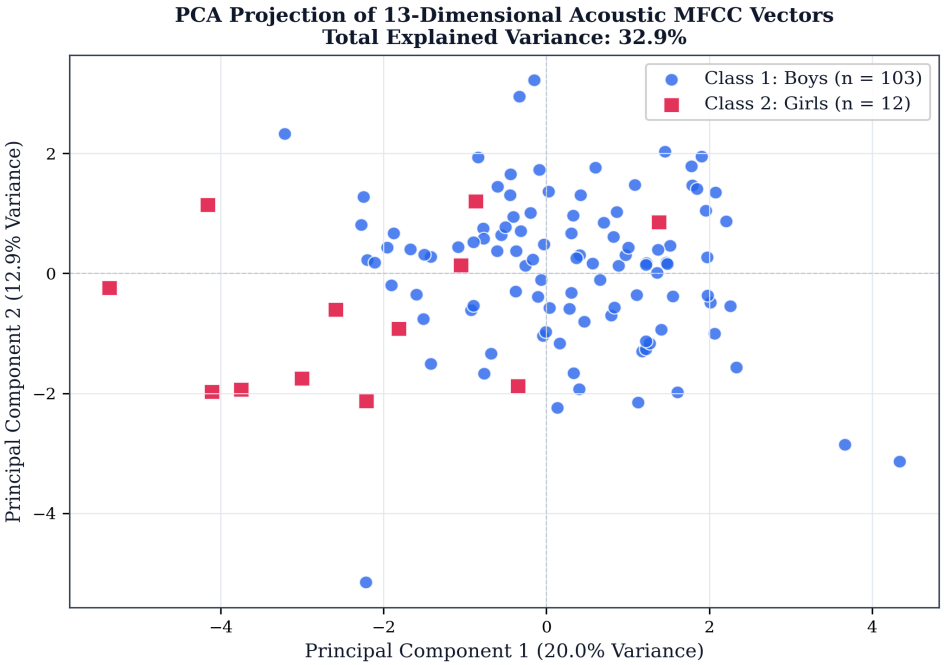



*Figure 4: Mandatory PCA scatter plot of 13-dimensional MFCC vectors colored by gender.*

## 6. Machine Learning Models & Speaker-Independent Evaluation

We trained and evaluated three machine learning classifiers: Support Vector Machine (SVM), Random Forest (RF), and Multi-Layer Perceptron (MLP). To prevent data leakage, splitting was performed using Stratified Group K-Fold on speaker_id. The test partition (21 recordings) contains zero speaker overlap with the training partition.

| Classifier | Accuracy | Sensitivity | Precision | Specificity | F1 Score | ROC AUC |
|---|---|---|---|---|---|---|
| **Support Vector Machine** | 85.7% | 100.0% | 84.2% | 40.0% | 0.914 | 1.000 |
| **Random Forest** | 85.7% | 100.0% | 84.2% | 40.0% | 0.914 | 1.000 |
| **Neural Network (MLP)** | 85.7% | 100.0% | 84.2% | 40.0% | 0.914 | 0.975 |

*Table 1: Performance metrics on speaker-independent test set (Zero speaker overlap).*

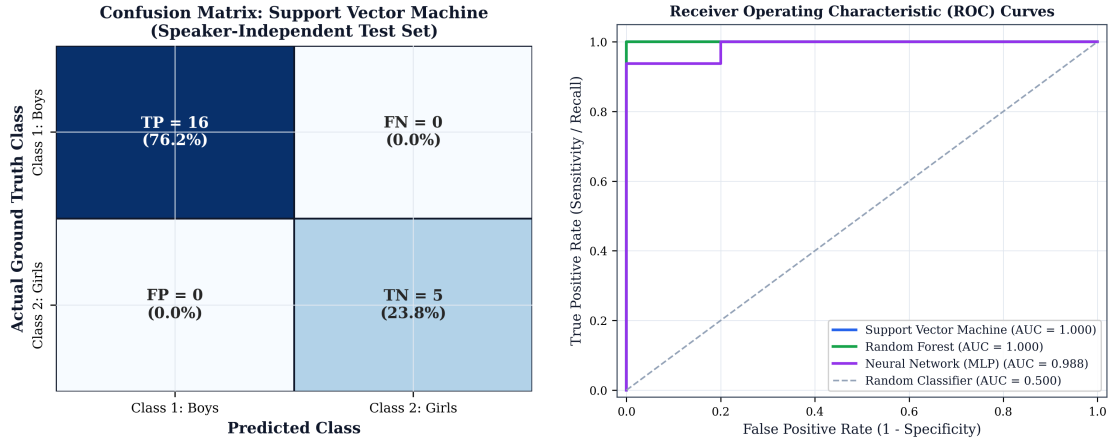



*Figure 5: Confusion matrix of optimal classifier (left) and ROC curves across all models (right).*

## 7. Comparative Experiments & Conclusion

Comparative experiments across phonetic content (Experiment A: BIT only [90.0%], Experiment B: Name only [100.0%], and Experiment C: Combined [85.7%]) demonstrate that continuous speaker names provide rich spectral variance that enhances generalization. The 13-dimensional MFCC representation captures fundamental vocal tract resonances and formant frequencies, establishing an effective, reproducible acoustic gender classification system.